Cane Water Audit: Systems and Site Parameters

This guide explains the six “Systems and Site” fields used in a water audit for a sugar factory and distillery, including definitions, data-entry guidance, and worked examples.

1. Explanation of Each Parameter

Parameter	Meaning	Unit	What to enter
Cooling tower cycles (measured COC)	Actual cycles of concentration, estimated as circulating-water conductivity divided by makeup-water conductivity under comparable conditions.	Dimensionless; e.g., 3.5	Use plant water-quality or conductivity records.
Cooling tower cycles (target COC)	Desired operating cycles that can be sustained without excessive scaling, corrosion, or fouling.	Dimensionless; e.g., 5.0	Use the target approved by engineering and water-treatment teams.
Major use points metered (%)	Percentage of identified major water-consuming locations fitted with functioning meters.	%	Calculate from the meter inventory and site inspection.
Annual rainfall	Average annual rainfall received at the factory site.	mm/year	Use reliable local meteorological records.
Roof and paved catchment	Effective roof and paved area draining into a rainwater collection or recharge system.	m²	Include only surfaces that can practically be connected.
Cost of fresh water	Effective cost of obtaining fresh water, including relevant purchase, pumping, treatment and supply costs.	₹/m³	Use actual factory costs; avoid double-counting.

2. Detailed Explanation and Examples

A. Cooling Tower Cycles of Concentration (COC)

Cooling towers reject heat by evaporating part of the circulating water. Dissolved minerals remain behind, increasing their concentration.

COC = Conductivity of circulating water ÷ Conductivity of makeup water

Example: makeup-water conductivity = 500 µS/cm; circulating-water conductivity = 1,750 µS/cm. Measured COC = 1,750 ÷ 500 = 3.5. If the target is 5.0, the plant may be able to reduce blowdown and makeup-water demand by increasing cycles, provided water chemistry, equipment limits and treatment controls permit it. A higher COC is not automatically better; the target must account for scaling, corrosion, fouling, tower design and the water-treatment programme.

B. Major Use Points Metered (%)

Typical points include raw-water intake, sugar-process makeup, boiler-feed and condensate systems, cooling-tower makeup and blowdown, imbibition water, distillery processes, effluent-treatment and reuse systems, domestic use, and green-belt irrigation.

Example: 22 of 40 identified major use points have functioning meters. Metering coverage = (22 ÷ 40) × 100 = 55%. Prioritise the main intake and the largest water-consuming processes.

C. Annual Rainfall and Rainwater Harvesting

Annual rainfall helps estimate the rainwater potentially available at the site. Consider seasonal variation, runoff losses and storage capacity.

Harvestable rainwater (m³/year) = Rainfall (mm) × Catchment area (m²) × Runoff coefficient ÷ 1,000

Example: rainfall = 700 mm/year; effective catchment = 10,000 m²; runoff coefficient = 0.80. Estimated harvestable volume = (700 × 10,000 × 0.80) ÷ 1,000 = 5,600 m³/year. This is a preliminary estimate before first-flush, filtration and storage losses.

D. Roof and Paved Catchment (m²)

Include roofs of factories, warehouses, workshops and suitable paved areas that can drain into the collection system. Do not automatically enter the total factory plot area. Roof runoff is often cleaner than industrial-yard runoff; paved areas may contain oil, grease, chemicals or suspended solids and may require separate treatment.

E. Cost of Fresh Water (₹/m³)

Include relevant purchase or tanker charges, municipal charges, raw-water pumping electricity and treatment costs. Use a consistent cost basis and do not double-count expenses already included in the water tariff.

Example: fresh-water cost = ₹30/m³ and water saved = 500 m³/day. Daily avoided cost = 500 × ₹30 = ₹15,000. For 300 operating days, estimated annual avoided cost = ₹45,00,000 (₹45 lakh), excluding other possible benefits.

3. Illustrative Inputs for an Initial Assessment

These values are illustrative assumptions only. Replace them with verified factory data before drawing conclusions.

Parameter	Illustrative input	Preferred data source
Measured COC	3.5	Cooling-tower operating records
Target COC	5.0	Engineering and water-treatment assessment
Major use points metered	55%	Meter inventory and site inspection
Annual rainfall	700 mm/year	Local meteorological records
Roof and paved catchment	10,000 m²	Site plan or measured area
Cost of fresh water	₹30/m³	Actual supply, pumping and treatment costs

4. How These Parameters Support Decisions

- Reduce water consumption through better metering, leak detection and cooling-tower management.
- Estimate rainwater-harvesting potential from rainfall, effective catchment area and runoff characteristics.
- Quantify the financial value of water conservation using the actual cost of fresh water.

Good practice: distinguish measured data from assumptions, document the source of each input, and validate water quality and operating constraints before implementing changes.